

StockGrid.

Next-Generation Inventory Management System

FULL PROJECT TECHNICAL OVERVIEW

Project Overview



Defining the future of stock visibility and predictive logistics.

| Vision and Core Purpose

Centralized Control

StockGrid is designed to bridge the gap between complex supply chains and user-friendly management. It provides a single source of truth for all inventory movements, ensuring data consistency across multiple departments.

Intelligent Scaling

Beyond simple tracking, StockGrid utilizes AI to offer predictive insights, helping businesses anticipate shortages and optimize their capital by reducing overstocking and dead inventory.

| The Modern Tech Stack



React & Vite

Built with React 18 for high-performance UI components and Vite for lightning-fast development and build cycles.



Node & Express

A robust backend architecture using Express.js to handle complex business logic, AI integrations, and real-time alerts.



Supabase

Leveraging PostgreSQL via Supabase for secure data storage, real-time subscriptions, and scalable authentication.

Frontend Experience

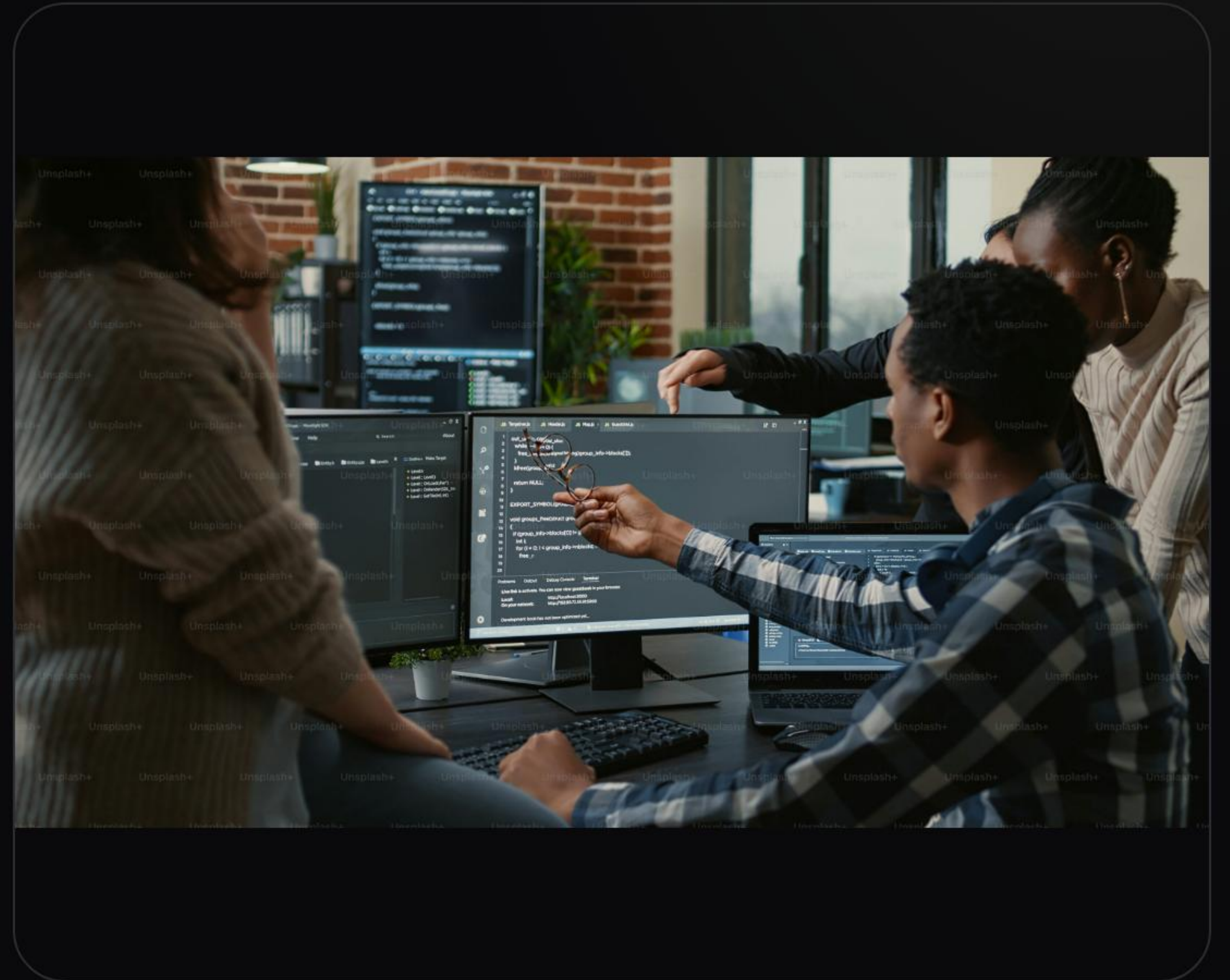
The StockGrid frontend is a masterpiece of modern UX/UI design. Developed with Tailwind CSS, it offers a fully responsive, dark-themed interface that prioritizes data readability and user efficiency.

Key features include dynamic dashboards, real-time stock bars, and interactive modals for rapid data entry and management.



| Robust Backend Logic

- 🔧 **AI Services:** Integrated AI service for generating predictive stock insights.
- 🔔 **Alert Engine:** Real-time alerting system for low stock and transaction anomalies.
- ✉️ **Email Integration:** Automated notification system using Nodemailer.
- 🛡️ **Auth Middleware:** Secure JWT-based routing and role verification.



| Performance Capabilities

100%

Real-time Visibility

Unmatched Precision

StockGrid eliminates the lag in inventory tracking. Every transaction, whether an import or a manual adjustment, is reflected system-wide in milliseconds, ensuring that the dashboard always shows the current state of your warehouse.

Database Architecture

Table Name	Primary Key	Foreign Keys	Purpose
Products	id (UUID)	supplier_id	Stores core product data and current stock levels.
Transactions	id (UUID)	product_id, user_id	Logs every stock movement (In/Out/Adjustment).
Suppliers	id (UUID)	-	Maintains contact and logistics info for vendors.
Profiles	id (UUID)	auth_user_id	Extended user information and role assignments.

Core Functionalities



Deep dive into the operational heart of StockGrid.

| AI-Powered Smart Insights



Predictive Analysis

Analyzes historical transaction data to predict future stock needs and seasonal trends.



Anomaly Detection

Identifies unusual stock movements that might indicate theft, leakage, or logging errors.



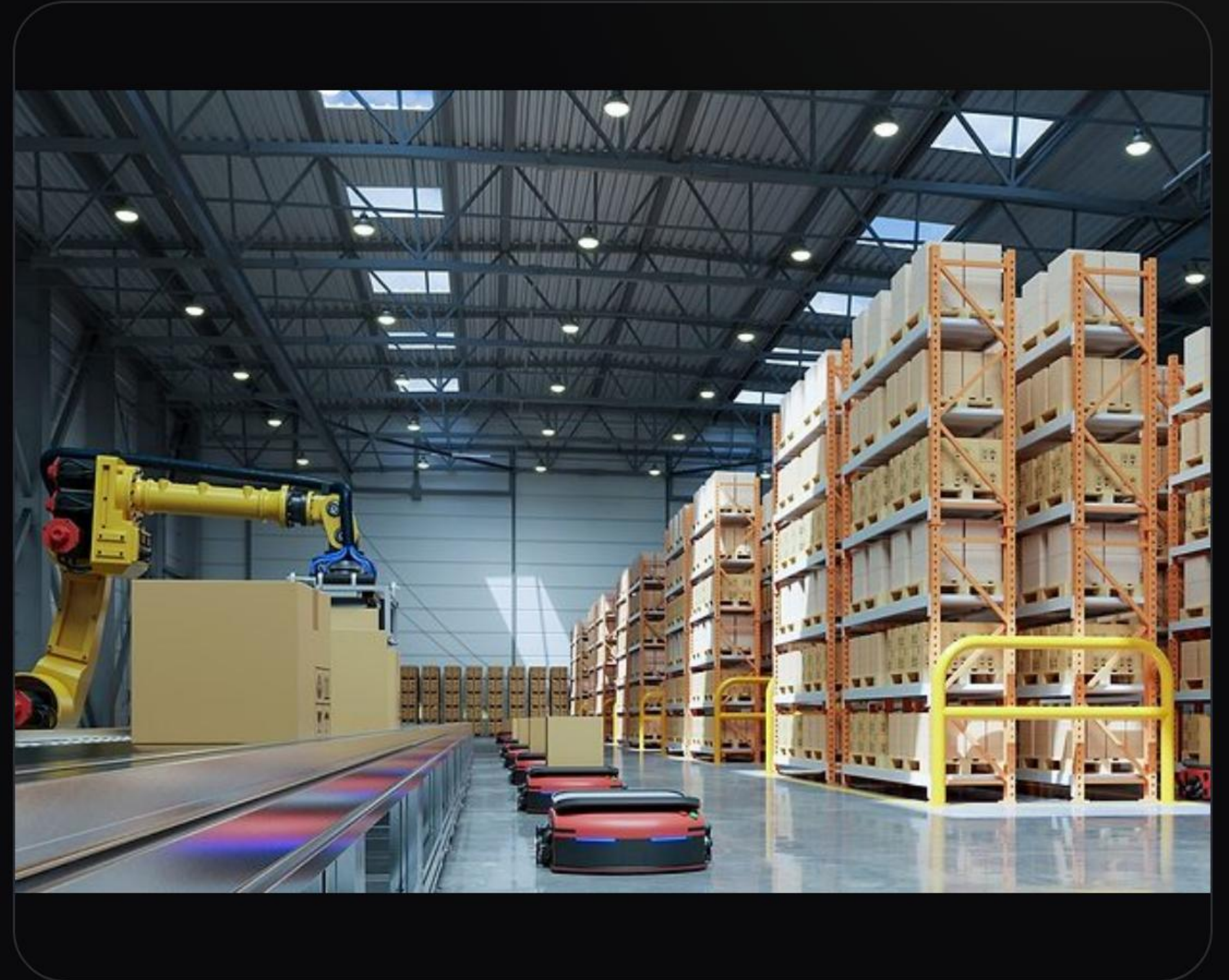
Smart Alerts

Automatically triggers low-stock notifications based on lead times and usage frequency.

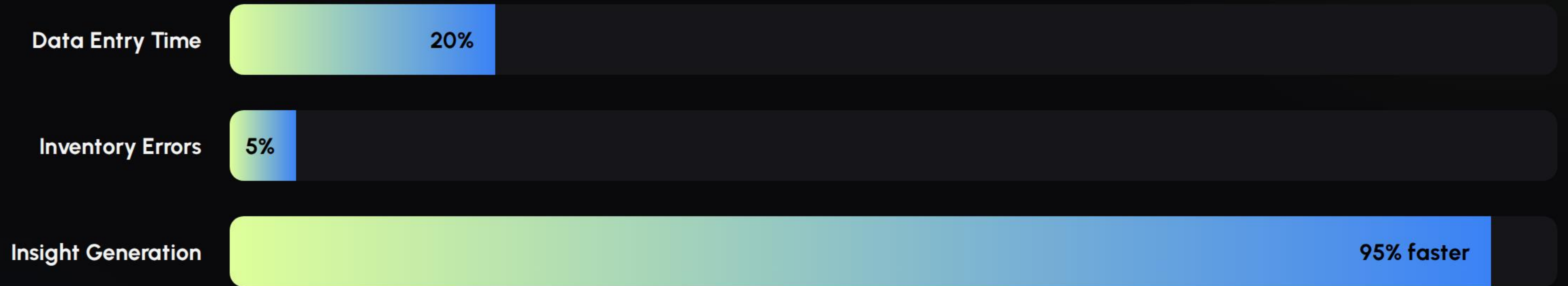
| Inventory & Transactions

StockGrid tracks the lifecycle of a product from initial procurement to final sale. Every interaction is logged with high precision.

- ✓ **Bulk Processing:** Upload 1000s of rows via CSV in seconds.
- ✓ **SKU Generation:** Intelligent auto-generation of unique SKUs.
- ✓ **Historical Audit:** Comprehensive logs for every stock change.



Manual vs. StockGrid Efficiency



*Comparison based on standardized logistics operational benchmarks.

| Security & Integrity

-  **End-to-End Encryption:** Sensitive data and passwords are hashed using Bcrypt before storage.
-  **Role-Based Access:** Granular permissions ensuring users only access what they need.
-  **Secure Password Reset:** Token-based reset system with unique verification links.
-  **Rate Limiting:** Protection against Brute-force and DDoS attacks via backend middleware.

Reporting Excellence

Data is useless without visualization. StockGrid generates comprehensive PDF and Excel reports for financial auditing and stock forecasting.

[SYSTEM LOG]: Report generated in 1.2s



Q&A.

Thank you for your attention. Any questions about the StockGrid implementation?

Repository: stockgrid-main

Version: 1.0.0 Stable

Image Sources



<https://cdn.dribbble.com/userupload/45969034/file/392b2a90c9e64a739e4966a45ff8ef4b.png?format=webp&resize=400x300&vertical=center>

Source: dribbble.com



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Source: unsplash.com



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Source: www.supplychainbrain.com



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